Banning a Vanishing Margin: Brazil's Child Marriage Ban, Formal Marriage, and Informal Unions*

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September 2026 — Preliminary

Abstract

In 2019 Brazil eliminated all legal pathways to civil marriage below age 16. We evaluate the reform using an age-eligibility design and three sources measuring distinct margins: civil-marriage registrations, coresident unions, and mothers' conjugal status at birth. The locked Registry model estimates a 1.1% decline at age 15 (95% CI $[-14.9\%, +14.9\%]$), whereas omitting age-specific trends yields -37.8%. A post-result rolling-origin protocol, frozen before its sensitivity estimates, finds that none of five candidate counterfactuals passes pre-2019 calibration; neither estimate is therefore design-wide. PNADC estimates a suggestive 0.40 percentage-point relative increase in coresident-union prevalence (95% CI $[-0.003, +0.801]$), but cannot establish informal substitution. SINASC's trended and no-trend estimates span -23.9% to $+29.3\%$ and fail placebo diagnostics. The evidence neither identifies an additional short-run fall in formal marriage nor establishes no effect. It shows why administrative declines, union formation, and legal formalization require separate measurement and credible counterfactual trends.

JEL: J12, J13, K36, O15, I38. **Keywords:** child marriage; minimum-age laws; informal unions; civil registration; Brazil.

*Working draft; please do not cite or circulate. The primary Registry and PNADC specifications were SHA-256–locked on 2026-09-02 before any post-reform coefficient was computed. Later extensions are separately dated and hash-locked; the Registry trend sensitivity is explicitly post-result. [ACKNOWLEDGMENTS TO BE ADDED]

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1 Introduction

Minimum-age-of-marriage laws are the standard policy instrument against child marriage. More than fifty countries have tightened them since 2000, and ending child marriage by 2030 is an explicit target of the Sustainable Development Goals. The evidence on whether these laws change behavior, however, is mixed at best: enforcement is weak in most of the developing world [Collin and Talbot, 2023], Mexico's state-level bans reduced *registered* child marriages while informal unions took their place [Bellés-Obrero and Lombardi, 2023], and an information experiment in Bangladesh found that publicizing a child-marriage reform could even backfire [Amirapu et al., 2026]. The core difficulty is that the law regulates a legal act—civil marriage—while the phenomenon of concern is a living arrangement that in much of the world requires no state involvement at all.

Brazil's Law 13,811 of March 2019 offers an unusually sharp version of this problem. The law closed the last legal pathways to civil marriage below age 16, which until then had been available in cases of pregnancy or to avoid the imposition or fulfillment of a criminal penalty. It did not touch marriages at ages 16–17, which remain lawful with parental consent. Two features make this reform a revealing case. First, treatment is defined by age alone and applies nationwide from a precise date, so an age-eligibility contrast is available in the universe of registration records. Second—and this is the substantive point—the margin the law banned was already close to extinction. In the pre-reform years 2013–2018, about 3,400 fifteen-year-olds married in the entire country, a rate of roughly 4 per 100,000; in the same period, household survey data put the share of 15-year-old girls living in a coresident union between 1.2% (a conservative definition) and 4.8% (a broad one). These quantities are not directly comparable—a registration rate is a flow and union prevalence is a stock—but together they show that the reform targeted a small formal margin alongside a larger observed stock of coresident unions. Vital records add a third measurement margin: among mothers younger than 16 giving birth in 2019, 0.7% declared themselves formally married, while 19% declared a *união estável*, the Brazilian informal-union status.

This paper asks what a national ban accomplishes in that environment. We evaluate Law 13,811/2019 with a research design frozen before estimation: the specification, treatment window, comparison groups, and inference procedure were locked—with cryptographic hashes—before any post-reform coefficient was computed, and every subsequent amendment is logged. The design is a difference-in-differences by age eligibility on the universe of civil marriage registrations from 2013 to 2024: persons aged 15 (directly treated) are compared with persons aged 17–19 (untreated), within region and quarter, in a Poisson pseudo-maximum-likelihood model with population offsets, region-by-age and region-by-period fixed effects, age-specific seasonality, and age-specific linear trends. Because the reform is a single national event, we do not treat regional cells as independent experiments: inference clusters on time periods and is triangulated with HAC-robust aggregate series, temporal bootstrap forecasts, placebo reform dates, and randomization inference, with all limitations reported.

Three results follow. First, the locked specification finds no detectable additional decline in marriage registrations at age 15 in the three fully treated quarters of 2019: the estimate is -1.1%, with a 95% confidence interval from -14.9% to $+14.9\%$, and an implied 2.1 registrations averted (CI $[-25.2, +33.8]$). This is an imprecise null, not evidence of a zero: the design's minimum detectable effect at 80% power is a 19.3% decline. Conditional on the locked linear-trend specification, the confidence interval excludes declines larger than 14.9%, but it does not do so across admissible counterfactual trends. The same model without age-specific trends yields -37.8% ($p < 0.001$). A post-result forecast-validation protocol, frozen before its own estimates, ranks the seasonal-level no-trend analogue last on pre-2019 prediction. The best-ranked global-linear forecast gives $+2.0\%$ (95% forecast interval $[-17.6\%, +23.4\%]$), and an inverse-RMSE^2 ensemble gives $+3.0\%$. Yet all five candidates fail the protocol's absolute calibration threshold. Their point estimates span $[-33.7\%, +12.8\%]$, a specification envelope rather than a confidence or causal bound; the seasonal-level model's own interval, $[-54.7\%, -0.3\%]$, includes a decline of 49%. The exercise therefore ranks extrapolations but cannot certify a counterfactual or rule out a large effect design-wide.

The closest benchmark in the literature, Mexico's staggered state bans, cut registered marriages of 16–17-year-old girls by 49 percent of the pre-reform mean [Bellés-Obrero and Lombardi, 2023]. That magnitude lies outside Brazil's locked-model interval, but the comparison is contextual rather than a cross-design hypothesis test. Mexico's significant effect arises at 16–17, where formal marriage was still common (roughly 1,700 registrations per 100,000 girl-years against Brazil's 4 per 100,000 at age 15),[1] while at 14–15—the margin comparable to ours—Bellés-Obrero and Lombardi [2023] likewise find no significant change. Read together, these age profiles are consistent with—but do not establish—the hypothesis that minimum-age laws have larger registration effects where formal marriage remains quantitatively important. We treat the failed calibration not as a robustness footnote but as a finding about how minimum-age laws can be mis-evaluated where the targeted margin is already declining.

Second, PNADC offers no evidence of a reduction in conservative coresident-union prevalence. In quarterly household survey data, the estimated prevalence among 15-year-olds does not fall after the reform; the point estimate is +0.399 percentage points (95% CI

[1]Derived from the published estimate: 0.695 fewer marriages per 1,000 girls per month over a 49 percent reduction implies a pre-reform mean of about 1.42 per 1,000 girl-months, or some 1,700 per 100,000 girl-years.

$[-0.003, +0.801]$, $p = 0.052$) relative to ages 17–19. The sign is consistent with substitution from formal marriage toward informal cohabitation—in Mexico, the authors report that, among adolescent mothers, the decline in the formally married share was "completely counteracted" by an increase in the informal-union share [Bellés-Obrero and Lombardi, 2023]—but we are explicit about what this coefficient can and cannot say: the registry measures a flow of formal events, the survey measures a stock of coresident arrangements, the two cannot be differenced, and equivalence testing shows the survey cannot rule out effects smaller than ± 0.5 p.p. The honest statement is that unions did not detectably decline, and the data lean toward relative informalization without establishing it.

Third, conjugal status at birth—observed for every live birth in the country, with the mother's exact date of birth and municipality of residence—provides a separate view of the small formal margin. Among mothers under 16, the formally married share fell from 1.7% in 2013 to 0.9% in 2016 and 0.7% in 2019; the monthly event study shows this share collapsing through 2016 and then running flat at roughly -33% relative to its pre-2017 level, with no break at the law. Precisely because the decay is convex, the pre-registered difference-in-differences with group-specific *linear* trends returns +29.3% (95% CI $[+11.3\%, +50.1\%]$; $p = 0.0008$). The positive estimate runs against the reform's direct eligibility mechanism, but the outcome is a stock conditional on giving birth, so composition can also move it. More decisively, the frozen diagnostics reject the design: the joint test of 24 leads has $p < 0.001$, and placebo reforms dated April 2017 and April 2018 "find" +19.7% and +36.8%. The no-trend variant returns -23.9%. We therefore report $[-23.9\%, +29.3\%]$ as a specification range, not a causal confidence interval. The fertility-composition contrast at age 15 is +1.6% (95% CI $[-1.3\%, +4.6\%]$; $p = 0.284$), and unknown-status reporting shows no detectable differential change ($p = 0.577$).

We make three contributions. To our knowledge, this is the first empirical evaluation centered on Brazil's 2019 ban,[2] complementing the Mexican staggered-adoption evidence [Bellés-Obrero and Lombardi, 2023] with a single-national-reform design and a triangulation across formal-flow, coresident-stock, and status-at-birth measures. Substantively, we document a limiting case with general interest: a ban that arrives after the formal margin has nearly disappeared primarily constrains formalization and may leave union prevalence unchanged; its evaluation is dominated by pre-trends. This sharpens the interpretation of

[2] A systematic search on September 2, 2026—nineteen logged queries across economics (SSRN, NBER, IZA, CEPR, ANPEC), demography, public health (SciELO, PubMed), Brazilian legal scholarship, and forward citations of the closest international studies—found no causal evaluation of Law 13,811/2019. Existing empirical work is descriptive: Urquia et al. [2022] explicitly present their 2011–2018 perinatal series as a baseline against which the 2019 reform could later be measured, and a student report supervised by one of the authors (Faria, 2025, Insper) documents descriptively that the decline predates the law.

enforcement findings in Collin and Talbot [2023] and of null-to-perverse legal effects in Amirapu et al. [2026]. Methodologically, we show how much discipline a frozen specification buys in single-reform settings: the gap between -1.1% and -37.8% shows how inference changes when pre-existing age-specific trends are modeled.

The rest of the paper proceeds as follows. Section 2 reconstructs the legal change from primary sources. Section 3 describes the three data sources and the measurement decisions, which in this setting carry identification weight. Section 4 presents the design, the specification lock, and inference for a single national reform. Section 5 reports registry results; Section 6 turns to unions and to status at birth; Section 7 collects robustness and threats. Section 8 discusses what the results imply for minimum-age laws and their monitoring.

2 Institutional background

Under the Civil Code of 2002, the marriageable age in Brazil is 18, or 16–17 with the authorization of both parents or legal representatives (art. 1.517) [Brasil, 2002]. Before 2019, art. 1.520 allowed marriage *below* 16 in two exceptional situations: to avoid the imposition or fulfillment of a criminal penalty (a clause tied to the pre-2005 treatment of sexual offenses) and in case of pregnancy. Law 13,811, published and in force on March 13, 2019, rewrote art. 1.520 to eliminate both exceptions: since then, no civil marriage below 16 can be lawfully celebrated in Brazil [Brasil, 2019].

Three institutional facts matter for the design. First, the law changed eligibility for a formal act performed before a civil registrar; it created no new enforcement apparatus directed at informal unions, and *união estável* remains a constitutionally recognized family form with no minimum-age registration requirement. Second, the treated population is precisely delimited by age at the date of celebration: persons under 16. Ages 16–17 are legally untouched, though behaviorally they may absorb postponed marriages, which is why our primary comparison group is ages 17–19 and contrasts involving 16–17 are interpreted as potentially contaminated by delay. Third, the reform was national and immediate, with no phase-in, no state discretion, and no grandfathering of pending authorizations that we could verify in primary sources; the identification problem is therefore the single-event problem, not the staggered-adoption problem, and we treat it as such throughout.

The registry data also show what the law was regulating. Marriages with at least one spouse below 16 fell from 1,093 in 2013 to 512 in 2018—before any reform—and to 419 in 2019 and 193 in 2024. The decline that matters for Brazilian adolescents was well under way for reasons the law did not cause, which is precisely why an evaluation that does not model it will misattribute it.

3 Data and measurement

Our three sources measure different objects. We keep them separate by construction and never subtract a stock coefficient from a flow coefficient.

3.1 Civil marriage registrations (flow of formal events)

The Civil Registry statistics (IBGE, SIDRA table 4406) record every civil marriage by single year of age of each spouse, sex, month, and place of *registration*, from 2013 to 2024. We reconstructed the official aggregates exactly: all 3,780 local age–sex cells and every verifiable local total match the published figures. Age refers to age at registration in completed years; the month is the month of registration, not celebration; and the geography is the registry office, not the couple's residence. Each of these measurement facts constrains the design: completed years rule out a conventional regression discontinuity, registration lags blur the treatment date (March 2019 is excluded; the post window opens in April/2019Q2), and office-versus-residence geography caps the defensible spatial resolution at the region level, per a precision rule fixed before estimation.

3.2 PNADC (stock of coresident unions)

The continuous national household survey (PNADC) provides quarterly cross-sections of 2.43 million adolescents aged 14–19 over 2012–2024 (48 quarters; calibrated weights; design-based variance using strata and PSUs). Our conservative union indicator equals one when the adolescent is the spouse/partner of the household head or is the head with a coresident partner; a broader indicator adds plausible nested pairings and is used only as robustness with an explicit ambiguity flag. The survey follows dwellings, not individuals; an adolescent who leaves home to form a union elsewhere may exit the sampled dwelling, so the indicator undercounts unions in a way that can correlate with the outcome. We therefore treat PNADC as measuring the stock of *coresident, household-grid-visible* unions, and we report equivalence tests and minimum detectable effects rather than interpreting non-significance as absence of effect.

3.3 SINASC (conjugal status at birth)

The national live-birth information system (SINASC, Ministry of Health) records every live birth with the mother's age—and, since the 2011 form, her exact date of birth—her declared conjugal status (single; married; widowed; separated/divorced; *união estável*; unknown), municipality of *residence*, and the child's date of birth, along with education, race, parity, and prenatal-care covariates. We use the official open microdata, 2013–2024 (2.4–2.9 million births per year; SHA-256-verified acquisition; layouts harmonized across the 2015/2016 file-generation break; audit in the replication package). Cross-validation against published SVSA totals confirmed the open files to the record for 2014 but revealed the 2015 export to be materially incomplete; 2015 is excluded from estimation under a logged data-error amendment (Section 6). Two caveats govern interpretation. Conjugal status is declared at the birth, without registry verification: it measures the stock of statuses among women giving birth, not a marriage flow, and post-law declarations could re-label rather than reflect behavior. And any SINASC estimand conditions on a birth having occurred, so it is a joint outcome of fertility and formalization; we present it as such. In exchange, SINASC observes 55–70 thousand births to mothers under 16 per year—against 194 treated-cell marriage registrations in 2019Q2–Q4—with residence geography and daily dates, which no other source offers.

3.4 Denominators and comparability

Population denominators by age, sex, region, and quarter come from PNADC with a cell-quality rule (minimum unweighted counts and maximum coefficients of variation) frozen before estimation; state-level cells failed the rule and region was selected as the primary geography. The under-15 registry category is reported as counts and shares only—no rate with a fabricated 0–14 population is ever computed.

4 Empirical strategy

4.1 Design and primary specification

Treatment is age eligibility: a 15-year-old could, in principle, marry under the pregnancy exception before March 13, 2019 and cannot since; a 17-year-old's legal regime is unchanged. The primary estimand is the short-run change in civil marriage registrations of 15-year-olds per 100,000 residents aged 15, in 2019Q2–Q4, relative to ages 17–19, in ratio and in level terms.

The locked model is a PPML regression on region × age × quarter counts (540 cells; 5 regions; 27 quarters; ages 15 and 17–19) with the log resident population as offset; region-by-age, region-by-period, and age-by-calendar-quarter fixed effects; age-specific linear trends; and the treatment indicator (age 15 × post). 2019Q1 is omitted (partial treatment); 2020–2021 is flagged as pandemic-contaminated and the primary window ends in 2019Q4. Alternative control sets (18–19; 16–17), no-trend and linear-rate variants, monthly frequency, and pre-window starts in 2014 and 2015 were all frozen as mandatory robustness before any estimate existed.

4.2 Inference for one national reform

The reform is a single national event: regional cells share one treatment date and do not constitute independent experiments. Primary inference clusters on time periods (27 clusters, 3 post). We triangulate with (i) an aggregate Brazil series with HAC standard errors, (ii) a temporal block bootstrap of the pre-period forecast, (iii) placebo reform dates in the pre-period, and (iv) randomization inference over feasible placebo dates, whose resolution with only four dates (p-floor = 1.00) we report rather than hide. A two-way cluster variant produced an implausibly small standard error after a positive-semidefinite adjustment with five regions; it is reported as a diagnostic and given no evidentiary weight.

4.3 The specification lock

The full specification—estimands, samples, windows, comparison groups, model, and inference—was frozen and hash-stamped before any post-reform coefficient was computed (2026-09-02T00:56, SHA-256 in the replication package). Four amendments, all logged before the corresponding re-estimation, concern collinearity in saturated event studies and one exact computational substitution; none was motivated by a sign, magnitude, or significance. The SINASC estimands entered through the same mechanism: a separate extension lock (v1.0.0) was frozen and hash-stamped on 2026-09-02, after the data audit but before any post-reform SINASC contrast was computed, with expected signs recorded ex ante, a disclosure of the descriptive quantities already seen, and a mandatory diagnostic battery—whose results are reported in Section 6 exactly because they reject the design.

5 Registry results

5.1 Raw series and the vanishing margin

Figure 1 plots registration rates by single age. The age-15 series is one to two orders of magnitude below ages 17–19 throughout and declines over the whole pre-period. In the 2013–2018 window, 3,394 fifteen-year-olds married nationwide (4.19 per 100,000 per year), against 94,107 sixteen-year-olds and 130,000 seventeen-year-olds. The under-15 category shrinks from 364 marriages in 2013 to 246 in 2016 and 85 in 2024.

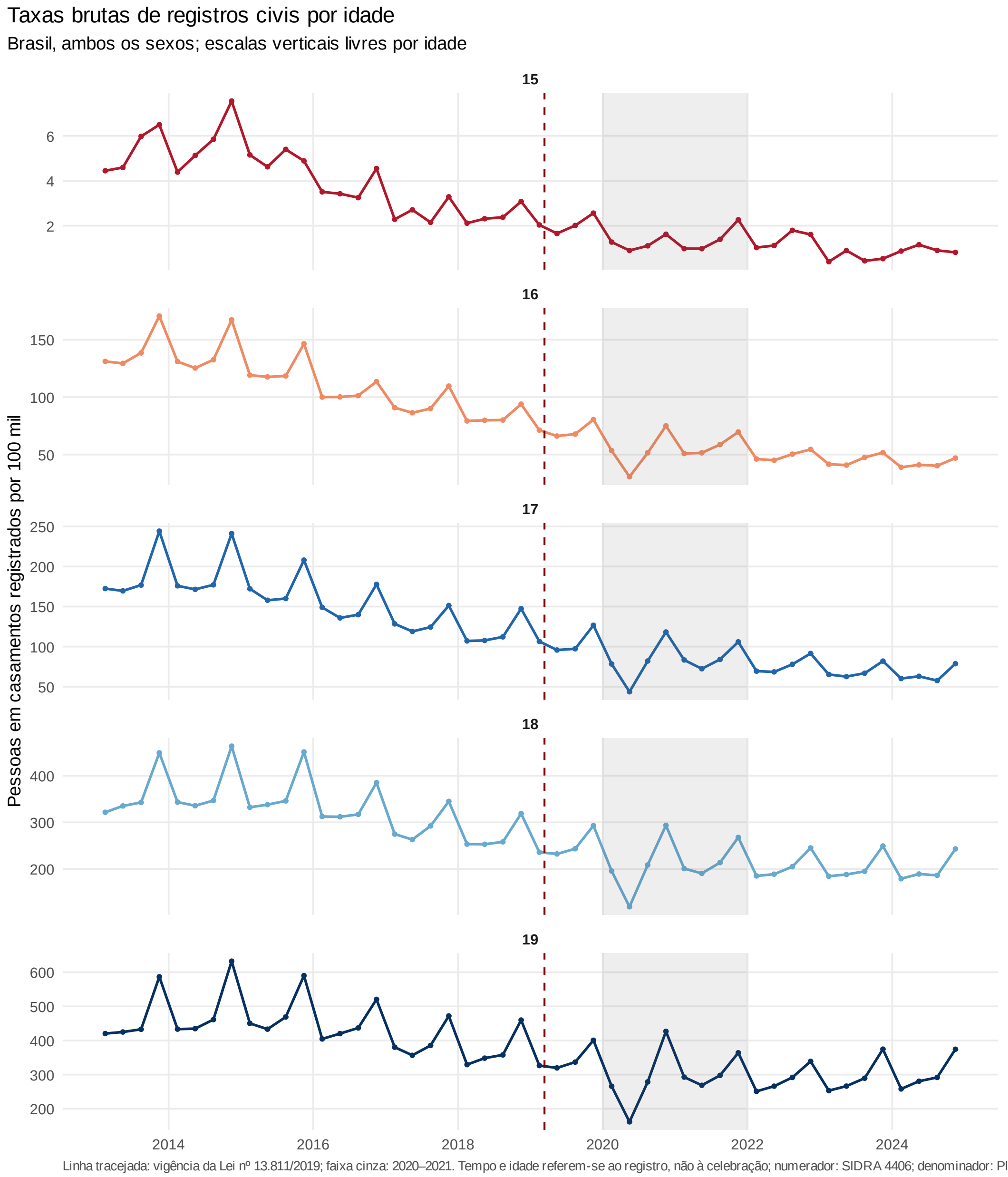

Figure 1: Civil marriage registrations per 100,000 residents, by single age at registration. Vertical line: Law 13,811 (March 2019). Source: SIDRA 4406 and PNADC denominators; replication package.

5.2 The locked estimate

The primary PPML contrast for 2019Q2–Q4 yields a rate ratio of 0.989: -1.1%, 95% CI $[-14.9\%, +14.9\%]$, $p = 0.888$ (period clusters). In levels this is -0.022 per 100,000, or 2.1 registrations averted against a counterfactual, with CI $[-25.2, +33.8]$. The aggregate HAC series gives $+2.0\%$ ($p = 0.785$) and the temporal bootstrap forecast $+2.0\%$ ($p = 0.857$). Figure 2 shows the event study: no post-reform break stands out from the pre-period variation.

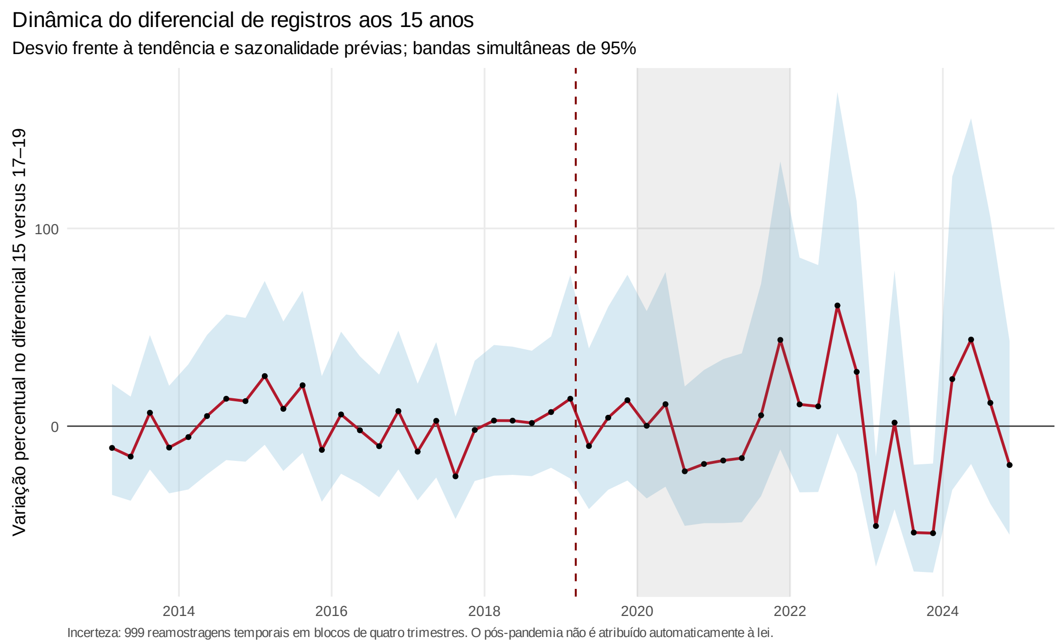

Figure 2: Event study, age 15 versus 17–19, locked specification. 2019Q1 omitted; post window 2019Q2–Q4. Source: replication package.

5.3 Trend dependence as a result

Removing the age-specific linear trends flips the estimate to -37.8% ($p < 0.001$). Monthly frequency with trends gives -2.4% ($p = 0.820$). Starting the pre-window in 2014 or 2015 moves the locked estimate to $+8.2\%$ and $+11.1\%$. A synthetic age control built on pre-2019 data places all weight on age 17 and shows a short-run gap of -0.658 per 100,000, which we read as descriptive corroboration, not new identification. The joint test of 24 leads does not reject ($p = 0.161$), but no simultaneous band stays within $\pm 10\%$, and one placebo date (2015Q2) is significant ($p = 0.002$): parallel evolution cannot be certified, it can only be modeled, and the conclusion depends materially on how. This is the central empirical lesson of the paper: in a collapsing series, the counterfactual trend *is* the estimate.

We therefore discipline the trend choice with a forecast-validation exercise, frozen as a post-result protocol before any of its sensitivity estimates were computed (five candidate counterfactuals for the national age-15 versus 17–19 log-rate gap; six overlapping three-quarter rolling windows over 2017Q1–2018Q2, fit strictly before each window; calibration tiers, ensemble weights, and bootstrap fixed in advance; Figure 3 and Appendix Table 4). The exercise disciplines in both directions. It discriminates: the seasonal-level candidate—the forecast analogue of the PPML no-trend specification—is decisively the worst forecaster (rolling RMSE 0.391 against 0.131 for the best; zero of six windows within a rate-ratio factor of 1.20). The four lower-RMSE trending models put the 2019Q2–Q4 deviation between +1.1% and +12.8%; their model-specific p-values are at least 0.41, and the inverse-RMSE^2 ensemble gives +3.0%. And it humbles: no candidate meets the absolute calibration bar fixed in the protocol (the best model's worst window still misses by a factor of 1.26), so the calibration gate *fails*. No cross-model range earns the name of confidence set—we report the point-estimate range $[-33.7\%, +12.8\%]$ as a specification envelope only. Moreover, the seasonal-level forecast interval $[-54.7\%, -0.3\%]$ includes the 49% Mexico benchmark. The same standard we apply to SINASC below therefore binds here symmetrically: the Registry counterfactual cannot be certified either. The difference between the two sources is one of degree, dictated by their diagnostics—four trending Registry models cluster near zero, while SINASC's placebo battery rejects its design outright—not one of asymmetric treatment.

5.4 Power, and the absence of recapture

The design's 80%-power MDE is a 19.3% decline (power is 28% against a 10% decline and 83% against 20%). These numbers acquire meaning against the literature's benchmark: the 49 percent drop in registered marriages that Mexico's bans produced at their bound margin [Bellés-Obrero and Lombardi, 2023] lies outside the locked-model confidence interval and is far beyond the region where that model has high power. But the MDE describes sampling precision conditional on the locked trend; it does not repair counterfactual identification. Because every forecast candidate fails calibration and the seasonal-level interval includes a 49% decline, the full exercise cannot rule out that benchmark design-wide. What the locked model cannot detect are effects below roughly one-fifth of the baseline, which at Brazil's age-15 rate of 4 per 100,000 correspond to a handful of marriages per year. Effects at ages 16–19 are +0.6%, +3.3%, +0.8%, and −2.4%, none surviving familywise correction; the aggregate "recapture" ratio (excess at 16–17 over deficit at 15) is undefined in half of the bootstrap draws and spans $[-114, +1{,}107]$ where defined. There is no reliable evidence of postponement toward legal ages, and with 2.1 point-estimated averted events there is little

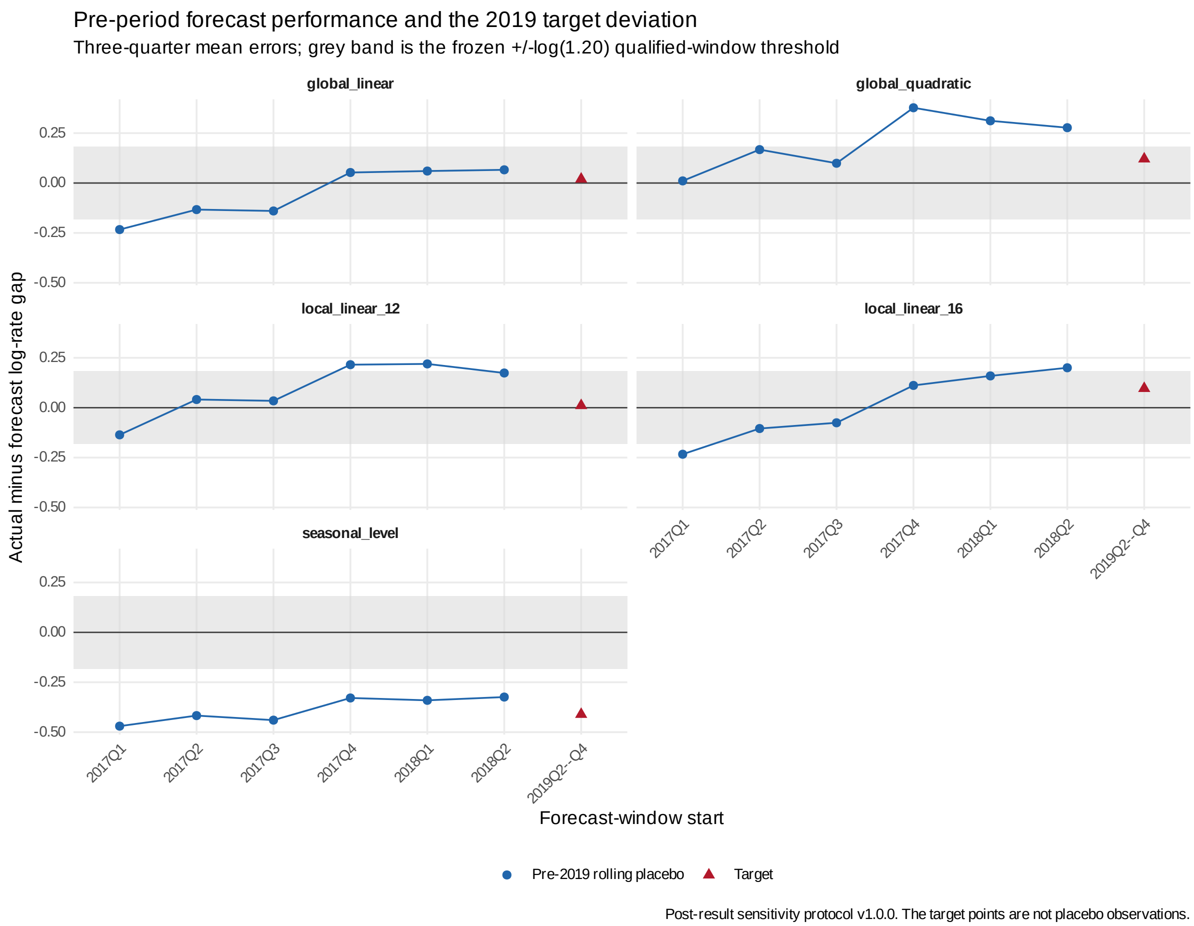

Figure 3: Rolling-origin validation of five counterfactual trend models. Blue circles are actual-minus-forecast mean log-rate gaps in six pre-2019 three-quarter windows; red triangles are the 2019Q2–Q4 target deviations. The gray band is the frozen $\pm \log(1.20)$ qualified-window threshold. Model fitting never uses the forecast window. Source: post-result sensitivity protocol and replication package.

to recapture. By sex, the female estimate is −9.7% (Holm $p = 0.125$); the male +155.8% is driven by 36 zero cells on a rare base and is not interpretable.

Table 1: Locked primary age-based PPML estimate, 2019Q2–Q4.

Model	Rate ratio (95% CI)	Change (95% CI)	Level effect per 100,000	Events averted (95% CI)	p
Locked PPML	0.989 [0.851, 1.149]	−1.1% [−14.9, 14.9]	−0.022	2.1 [−25.2, 33.8]	0.888

Notes: Region × age × quarter PPML with population offset; ages 17–19 are controls. Inference clusters by period (27 periods; five regions). The confidence interval for events averted reverses the sign of the corresponding event-effect interval.

6 Unions and formalization at birth

6.1 Coresident unions (PNADC)

The point estimate for the conservative union indicator among 15-year-olds is positive after the reform. The design-based cell estimate is +0.399 p.p. (95% CI $[-0.003, +0.801]$, $p = 0.052$); the aggregate HAC series gives +0.235 p.p.; microdata LPMs with household-period clustering or full survey-design variance give +0.344 p.p. (CIs $[-0.119, +0.807]$ and $[-0.025, +0.712]$); the broad indicator gives +0.597 p.p. ($p = 0.102$). The MDE is 0.575 p.p. and equivalence at ±0.50 p.p. fails (TOST $p = 0.311$). Estimates shrink as the pre-window shortens (0.399, 0.329, 0.113 p.p. for windows starting 2013, 2014, 2015). We conclude: no evidence of a reduction in conservative coresident-union prevalence; a positive, window-sensitive signal consistent with relative informalization that the survey cannot establish at conventional strength.

6.2 Conjugal status at birth (SINASC)

Among the roughly 55–80 thousand mothers under 16 giving birth each year, the formally married share was already small and falling before the law: 1.72% in 2013 (1,345 of 78,062), 0.88% in 2016 (593 of 67,419), and 0.69% in 2019 (374 of 54,166), against informal-union shares of 27.9%, 23.2%, and 19.5%. Mothers aged 16–17 in 2019: 3.9% married, 23.3% in *união estável*. Monthly counts of married under-16 mothers in 2019 (19–47 per month) show no visible April collapse, consistent with status-at-birth being a slowly decaying stock among parturients rather than a marriage flow. Validation against the official SVSA epidemiological bulletin uncovered a material defect in the open microdata: the 2014 file matches the published total to the record (2,979,259), but the 2015 export is short by an implied 231,143

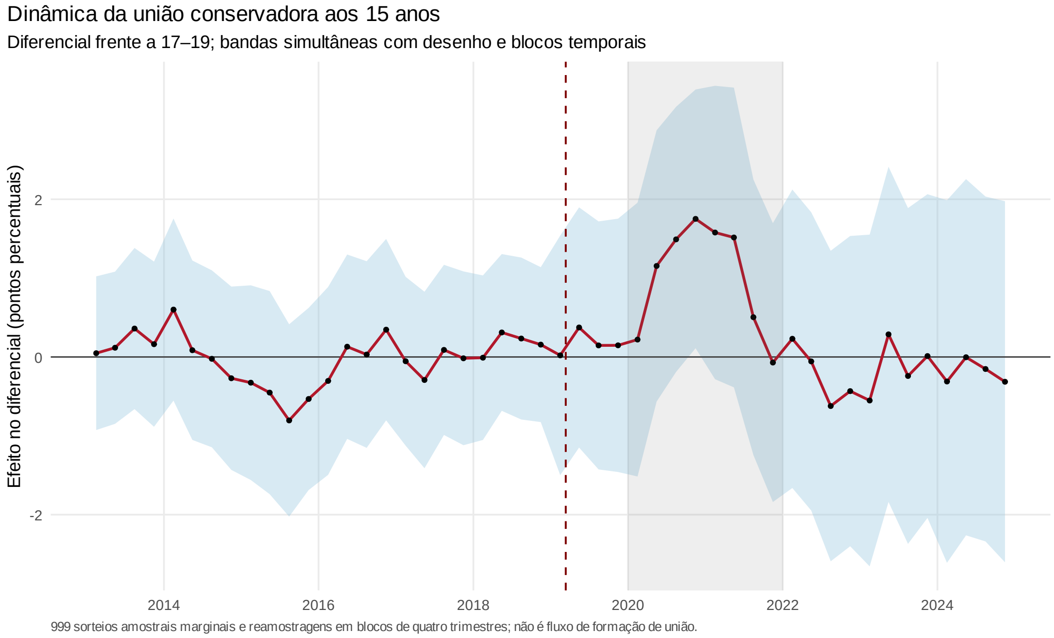

Figure 4: Coresident-union prevalence at age 15 (conservative definition), event study relative to ages 17–19. Source: replication package.

records (7.7%), concentrated in October–December; calendar year 2015 is therefore excluded from every estimate below, under a data-error amendment logged before re-estimation.

The pre-registered contrast (extension lock v1.0.0: mothers 10–15 versus 17–19, region-of-residence by month PPML on counts of married births with valid-status births as exposure, the registry's fixed-effect and trend structure, period clustering) returns +29.3% (95% CI $[+11.3\%, +50.1\%]$, $p < 0.001$; 270 married births in the treated post cells; Appendix Tables 2–3). We recorded the expected sign as negative-or-null before estimation. The positive estimate runs against the direct eligibility mechanism, while conditioning on birth leaves room for composition; the frozen diagnostics decide whether the contrast is credible, and they reject it. The monthly event study (Figure 5) shows the treated married share falling steeply through 2016 and then running flat at -30 to -37% of its pre-2017 level, straight across the reform date; the joint test of 24 leads has $p \approx 0$; and placebo bans dated April 2016, 2017, and 2018 return -5.9% (n.s.), $+19.7\%$ ($p = 0.004$), and $+36.8\%$ ($p < 0.001$)—a monotone-in-time pattern that is the signature of a linear trend extrapolating a convex, floor-approaching decay, not of treatment. The no-trend variant returns -23.9%; every trended variant (alternative controls, treated age 15 only, exact maternal age from dates of birth, quarterly aggregation, missing-as-unmarried) sits between +22% and +33%—except the variant whose pre-window starts in 2016, after the flattening, which returns +1.5% ($p = 0.86$), exactly as the artifact reading predicts. We therefore report $[-23.9\%, +29.3\%]$

as a specification range, not a causal confidence interval. The *união-estável* and any-union contrasts (+3.7% and +5.1%) inherit the same defect and receive no causal reading. Two frozen diagnostics survive: the fertility-composition contrast—births to mothers aged 15 per 100,000 girls aged 15—is +1.6% (95% CI $[-1.3\%, +4.6\%]$; $p = 0.284$), and unknown-status reporting shows no detectable differential change ($p = 0.577$). What SINASC establishes, then, is descriptive and disciplinary at once: the formal margin at birth was small and had stopped falling *before* the ban, and even a universe-sized administrative dataset cannot rescue a short-run contrast whose counterfactual trend is the conclusion.

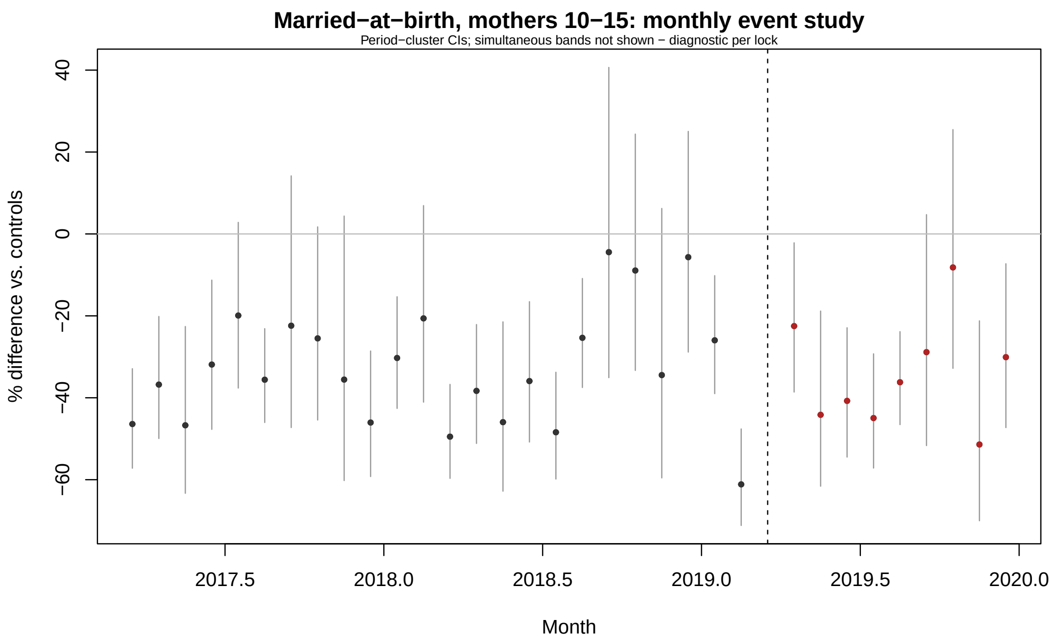

Figure 5: Married-at-birth among mothers 10–15 versus 17–19: monthly event study relative to the pooled pre-2017 baseline. Period-cluster confidence intervals; diagnostic per the extension lock. Source: replication package.

7 Robustness and threats to identification

The threats we can address are addressed by design: registration lag (omit March/Q1; monthly variant), pandemic contamination (primary window ends 2019Q4; 2020–2021 flagged), geographic mismatch (region-level analysis; municipal contrasts blocked), denominator uncertainty (repeated draws; SD an order below the temporal SE), and multiplicity (Holm within families). A pre-exposure triple-difference—regions with higher pre-law shares of under-16 marriages—shows a positive but imprecise gradient (+9.9% per SD, $p = 0.332$) and a -0.618 holdout correlation between exposure and 2018 changes, indicating mean reversion; it does not identify the national effect and is reported as descriptive. The threats we cannot address are named rather than assumed away: anticipation between the bill's passage and enforcement; age measured in completed years; nationally synchronized age-specific shocks, which no fixed effect absorbs; and the selection inherent in any status-at-birth outcome.

8 Discussion and conclusion

Brazil banned civil marriage below 16 after registrations at that age had become rare, while coresident unions remained more prevalent on a conceptually different stock measure. The locked Registry model finds no detectable additional decline in the banned formal events, but the post-result rolling-origin gate rejects every candidate counterfactual; the paper therefore establishes neither an additional short-run decline nor its absence. PNADC finds no detectable reduction in conservative coresident-union prevalence, with a suggestive—unproven—tilt toward informality. The status-at-birth margin had flattened before the reform, and its pre-registered contrast fails the placebo battery, leaving a specification range that spans zero.

Two implications travel beyond Brazil. For policy, minimum-age laws are instruments aimed at the recorded margin; where unions are informal, the recorded margin can go to zero while the phenomenon continues, and the case for complementary instruments (education, income support, enforcement against coerced unions) is correspondingly stronger. Read jointly, the Mexican and Brazilian estimates are consistent with a gradient by the size of the formal margin: Mexico's bans reduced registrations sharply at ages 16–17, while neither Mexico at ages 14–15 nor Brazil below 16 identifies a comparable fall; informal-union measures did not decline commensurately [Bellés-Obrero and Lombardi, 2023]. This two-country comparison is suggestive, not a general causal law, because ages, outcomes, and designs differ. For evaluation and monitoring, the gap between our locked -1.1% and the naive -37.8% is a warning: registered child marriages were falling secularly in many countries before their bans, and legal-impact studies that do not model that trajectory can attribute secular decline to the law. Out-of-sample forecast validation ranks the no-trend candidate worst, but its failure to validate any candidate also prevents us from elevating the locked model into a design-wide estimate. The status-at-birth exercise shows the mirror image—a mis-specified trend can just as easily manufacture a *positive* "effect" of a ban, and only a pre-committed placebo battery tells the two apart—a bias that survey-based monitoring of the SDG target, which measures unions rather than marriages, will not correct in the opposite direction. Honest evaluation in this literature requires pre-committed specifications, explicit power statements, and the discipline of keeping flows, stocks, and statuses apart.

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A SINASC status-at-birth exhibits

Table 2: SINASC status-at-birth contrasts, mothers 10–15 versus 17–19, 2019M4–M12 (extension lock v1.0.0).

Outcome	Trended effect (95% CI)	p (Holm)	No-trend effect	MDE_{80}	TOST p	Treated events
Married (S1)	29.3% [11.3, 50.1]	0.002	−23.9%	23.8%	0.892	270
União estável (S2)	3.7% [0.9, 6.7]	0.010	−3.7%	4.1%	< 0.001	7,848
Any declared union (S3)	5.1% [2.2, 8.0]	0.001	−3.7%	4.0%	< 0.001	8,118

Notes: PPML on region-of-residence × group × month cells; the offset is births with valid status. Inference clusters by period. Column 2 uses group-specific linear trends; column 4 omits them. The TOST equivalence interval for the rate ratio is [0.85, 1.176]. Frozen diagnostics reject a causal reading of all three contrasts (Table 3); the table documents specification dependence.

Table 3: SINASC frozen diagnostics: placebo reform dates, lead test, aggregate HAC, misreporting, and fertility-composition contrasts.

Object	% effect	95% CI	p
Placebo ban April 2016	-5.9	[-21.1; 12.2]	0.497
Placebo ban April 2017	19.7	[5.9; 35.3]	0.004
Placebo ban April 2018	36.8	[16.6; 60.6]	<0.001
Brazil monthly gap, HAC(12)	24.8	[3.5; 50.4]	0.024
Unknown-status share (misreporting)	3.4	[-8.1; 16.3]	0.577
Fertility, age 15 (S4)	1.6	[-1.3; 4.6]	0.284
Fertility, age 14 (S4 rob.)	-3.0	[-6.1; 0.2]	0.137
Joint test, 24 monthly leads		$p < 0.001$	

Placebo bans use pre-2019 data only, with the true model and windows transposed to each placebo year. Fertility rows: PPML per 100,000 female residents of the same age (PNADC denominators), quarterly, Holm within family.

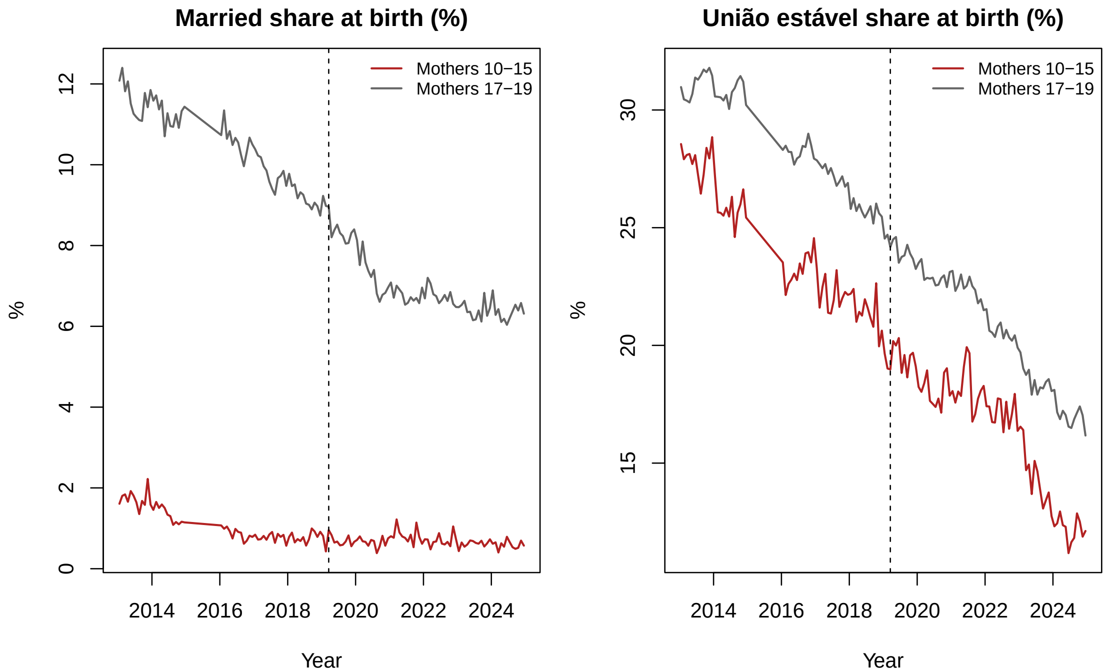

Figure 6: Declared conjugal status at birth, Brazil, monthly 2013–2024: married and *união estável* shares among mothers aged 10–15 and 17–19. Vertical line: Law 13,811 (March 2019). Source: replication package.

B Registry and PNADC exhibits

Table 4: Post-result forecast validation of counterfactual trend models (frozen lock v1.0.0). Calibration gate: *failed* — no candidate reaches the qualified tier; the locked PPML remains primary and the cross-model range is a specification envelope, not a confidence set.

Candidate	RMSE	Max |err|	≤1.20	Tier	2019Q2–Q4 %	Forecast 95%	p
Global linear trend	0.131	0.233	5/6	fail	2.0	[-17.6; 23.4]	0.85
Local linear (12q)	0.156	0.219	4/6	fail	1.1	[-14.4; 18.8]	0.89
Local linear (16q)	0.157	0.233	4/6	fail	10.1	[-10.8; 37.0]	0.42
Global quadratic	0.243	0.377	3/6	fail	12.8	[-13.6; 48.3]	0.41
Seasonal level (no trend)	0.391	0.470	0/6	fail	-33.7	[-54.7; -0.3]	0.04

Six overlapping three-quarter rolling windows, 2017Q1–2018Q2, models fit strictly before each window; RMSE of window-mean errors (log scale). Intervals: circular moving-block bootstrap (1,999 draws) of the three-quarter mean forecast error. Inverse-RMSE^2 ensemble: +3.0% (point only). All-model envelope: $[-33.7\%, +12.8\%]$.

Table 5: Trend and control-group robustness.

Specification	Controls	$\widehat{\beta}$ (SE)	p	Rate ratio	Change
Primary: age trends	17–19	−0.011 (0.077)	0.888	0.989	−1.1%
No age trends	17–19	−0.476 (0.073)	< 0.001	0.622	−37.8%
Age trends, alternative controls	18–19	−0.003 (0.079)	0.971	0.997	−0.3%
Age trends, potentially contaminated controls	16–17	−0.033 (0.070)	0.637	0.967	−3.3%

Notes: Period-cluster standard errors. The coefficient is on the log-rate-ratio scale; "Change" is $100[\exp(\widehat{\beta}) - 1]$.

Table 6: Age-specific event effects and aggregate recapture.

Age	Controls	Change	Estimated event effect (95% CI)	Aggregate recapture
15	17–19	−1.1%	−2.1 [−48.3, 31.8]	174.8
16	18–19	0.6%	40.6 [−457.7, 518.1]	—
17	18–19	3.3%	326.9 [−135.3, 781.0]	—

Notes: Event effects are fitted counts with treatment minus fitted counts without treatment, so a negative age-15 value corresponds to estimated events averted. The aggregate recapture ratio is unstable, does not track people, and is undefined in half of the bootstrap draws; it receives no causal weight.

Table 7: Primary design-based estimate for conservative coresident union.

Effect (p.p.)	95% CI	p	MDE_{80} (p.p.)	TOST p	Unweighted N (cases)
0.399	[−0.003, 0.801]	0.052	0.575	0.311	1,058,334 (57,141)

Notes: Combined standard error is 0.205 percentage points, obtained by quadrature of temporal and marginal design variance. TOST margin is ±0.5 p.p.; equivalence at 5% is not established.

Table 8: Inference triangulation for the national reform.

Method	Scale	Estimate (SE)	95% CI	p	Caveat
PPML, period clusters (primary)	Log rate ratio	−0.011 (0.077)	[−0.161, 0.139]	0.888	Only three treated post-period clusters.
PPML, region and period clusters	Log rate ratio	−0.011 (0.0004)	[−0.0115, −0.0101]	< 0.001	Five region clusters; covariance required a positive-semidefinite adjustment. Not conclusion-supporting.
PPML, region clusters	Log rate ratio	−0.011 (0.089)	[−0.186, 0.164]	0.904	Five clusters; diagnostic only.
Brazil age-15/control gap, Newey–West(4)	Aggregate log-rate gap	0.020 (0.073)	[−0.123, 0.162]	0.785	National-series approximation.
Pre-period linear-seasonal forecast, moving-block bootstrap	Forecast log-rate gap	0.020 (0.103)	[−0.185, 0.215]	0.857	Pre-period residual block bootstrap.
Prespecified-date randomization inference	Studentized PPML statistic	−0.141	—	1.000	Only four locked pseudo-dates; p-values are coarse.

Notes: Estimand scales differ as labeled. Diagnostics with inadequate cluster support are reported but receive no conclusion-supporting weight.

Table 9: Minimum detectable effects and power diagnostics.

Outcome	Target	Approx. power	SE	MDE_{80}	Caveat
Formal marriage, age 15	−10%	0.280	0.077	−19.3%	Normal approximation; period-cluster SE and three treated post quarters.
Formal marriage, age 15	−20%	0.830	0.077	−19.3%	Same calculation.
Formal marriage, age 15	−30%	0.997	0.077	−19.3%	Same calculation.
Formal marriage, age 15	−40%	1.000	0.077	−19.3%	Same calculation.
Conservative union, age 15	—	0.800	0.0021	0.575 p.p.	Temporal plus marginal design variance.

Notes: Two-sided $\alpha = 0.05$. Registry MDE is expressed as a decline; the union MDE is in percentage points.